$T^{-1}(x_1, x_2, x_3) = (y_1, y_2, y_3)$

$\Rightarrow T(y_1, y_2, y_3) = (x_1, x_2, x_3)$

$\Rightarrow T(y_1 e_1 + y_2 e_2 + y_3 e_3) = (x_1, x_2, x_3)$

$\Rightarrow y_1 T(e_1) + y_2 T(e_2) + y_3 T(e_3) = (x_1, x_2, x_3)$

$\Rightarrow y_1(e_1+e_2) + y_2(e_2+e_3) + y_3(e_1+e_2+e_3) = (x_1, x_2, x_3)$

$\Rightarrow (y_1+y_3,\ y_1+y_2+y_3,\ y_2+y_3) = (x_1, x_2, x_3)$

$\Rightarrow x_1 = y_1 + y_3$

$x_2 = y_1 + y_2 + y_3$

$x_3 = y_2 + y_3$

$\Rightarrow y_2 = x_2 - x_1$

$y_1 = x_2 - x_3$

$y_3 = x_1 - x_2 + x_3$

$\therefore T^{-1}(x_1, x_2, x_3) = (x_2 - x_3,\ x_2 - x_1,\ x_1 - x_2 + x_3)$

HW Thm-4.5.1 (Page-127)

HW $T : U \longrightarrow V$

$L(U, V)$ = Set of all Linear transformation is a V.S. (Prove it)

P Kalika Maths